

A Technical Introduction to DenseNet

Section 1

During the late 1980s, "skip-layer" connections were sometimes used in neural networks. Examples include: Lang and Witbrock (1988) trained a fully connected feedforward network where each layer skip-connects to all subsequent layers, like the later DenseNet (2016). In this work, the residual connection was the form $x \mapsto F(x) + P(x)$, where P is a randomly-initialized projection connection. They termed it a "short-cut connection". An early neural language model used residual connections and named them "direct connections".

Section 2

Degradation problem Sepp Hochreiter discovered the vanishing gradient problem in 1991 and argued that it explained why the then-prevalent forms of recurrent neural networks did not work for long sequences. He and Schmidhuber later designed the LSTM architecture to solve this problem, which has a "cell state" c_t that can function as a generalized residual connection. The highway network (2015) applied the idea of an LSTM unfolded in time to feedforward neural networks, resulting in the highway network. ResNet is equivalent to an open-gated highway network.

Section 3

Projection connection If the function F is of type $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ where $n \neq m$, then $F(x) + x$ is undefined. To handle this special case, a projection connection is used:

Section 4

Pre-activation block The pre-activation residual block applies activation functions before applying the residual function F . Formally, the computation of a pre-activation residual block can be written as:

Section 5

$$\frac{\partial E}{\partial x} = \frac{\partial E}{\partial x_L} \frac{\partial x_L}{\partial x} = \frac{\partial E}{\partial x_L} (1 + \frac{\partial}{\partial x} \sum_{i=1}^L F(x_i)) = \frac{\partial E}{\partial x_L} + \frac{\partial E}{\partial x_L} \frac{\partial}{\partial x} \sum_{i=1}^L F(x_i) = \frac{\partial E}{\partial x_L} + \frac{\partial E}{\partial x_L} \sum_{i=1}^L \frac{\partial F(x_i)}{\partial x} = \frac{\partial E}{\partial x_L} + \frac{\partial E}{\partial x_L} \sum_{i=1}^L \frac{\partial F(x_i)}{\partial x_L} \frac{\partial x_L}{\partial x} = \frac{\partial E}{\partial x_L} (1 + \sum_{i=1}^L \frac{\partial F(x_i)}{\partial x_L})$$

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This formulation suggests that the gradient computation of a shallower layer, $\frac{\partial E}{\partial x} = \frac{\partial E}{\partial x_L} (1 + \sum_{i=1}^L \frac{\partial F(x_i)}{\partial x_L})$, always has a later term $\frac{\partial E}{\partial x_L} \sum_{i=1}^L \frac{\partial F(x_i)}{\partial x_L}$ that is directly added. Even if the gradients of the $F(x_i)$ terms are small, the total gradient $\frac{\partial E}{\partial x}$ resists vanishing due to the added term $\frac{\partial E}{\partial x_L} \sum_{i=1}^L \frac{\partial F(x_i)}{\partial x_L}$.

Section 7

Previous work Residual connections were noticed in neuroanatomy, such as Lorente de No (1938). McCulloch and Pitts (1943) proposed artificial neural networks and considered those with residual connections. In 1961, Frank Rosenblatt described a three-layer multilayer perceptron (MLP) model with skip connections. The model was referred to as a "cross-coupled system", and the skip connections were forms of cross-coupled connections.

Section 8

where P is typically a linear projection, defined by $P(x) = Mx$ where M is a $m \times n$ matrix. The matrix is trained via backpropagation, as is any other parameter of the model.

Section 9

Stochastic depth is a regularization method that randomly drops a subset of layers and lets the signal propagate through the identity skip connections. Also known as DropPath, this regularizes training for deep models, such as vision transformers. ResNeXt (2017) combines the Inception module with ResNet. Squeeze-and-Excitation Networks (2018) added squeeze-and-excitation (SE) modules to ResNet. An SE module is applied after a convolution, and takes a tensor of shape $R^H \times W \times C$ (height, width, channels) as input. Each channel is averaged, resulting in a vector of shape R^C . This is then passed through a multilayer perceptron (with an architecture such as linear-ReLU-linear-sigmoid) before it is multiplied with the original tensor. It won the ILSVRC in 2017.

Section 10

Applications Originally, ResNet was designed for computer vision. All transformer architectures include residual connections. Indeed, very deep transformers cannot be trained without them. The original ResNet paper made no claim on being inspired by biological systems. However, later research has related ResNet to biologically-plausible algorithms. A study published in Science in 2023 disclosed

the complete connectome of an insect brain (specifically that of a fruit fly larva). This study discovered "multilayer shortcuts" that resemble the skip connections in artificial neural networks, including ResNets.

Section 11

A residual neural network (also referred to as a residual network or ResNet) is a deep learning architecture in which the layers learn residual functions with reference to the layer inputs. It was developed in 2015 for image recognition, and won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) of that year. As a point of terminology, "residual connection" refers to the specific architectural motif of $x \mapsto f(x) + x$, where f is an arbitrary neural network module. The motif had been used previously (see §History for details). However, the publication of ResNet made it widely popular for feedforward networks, appearing in neural networks that are seemingly unrelated to ResNet. The residual connection stabilizes the training and convergence of deep neural networks with hundreds of layers, and is a common motif in deep neural networks, such as transformer models (e.g., BERT, and GPT models such as ChatGPT), the AlphaGo Zero system, the AlphaStar system, and the AlphaFold system.

Section 12

Subsequent work Wide Residual Network (2016) found that using more channels and fewer layers than the original ResNet improves performance and GPU-computational efficiency, and that a block with two 3×3 convolutions is superior to other configurations of convolution blocks. DenseNet (2016) connects the output of each layer to the input to each subsequent layer: